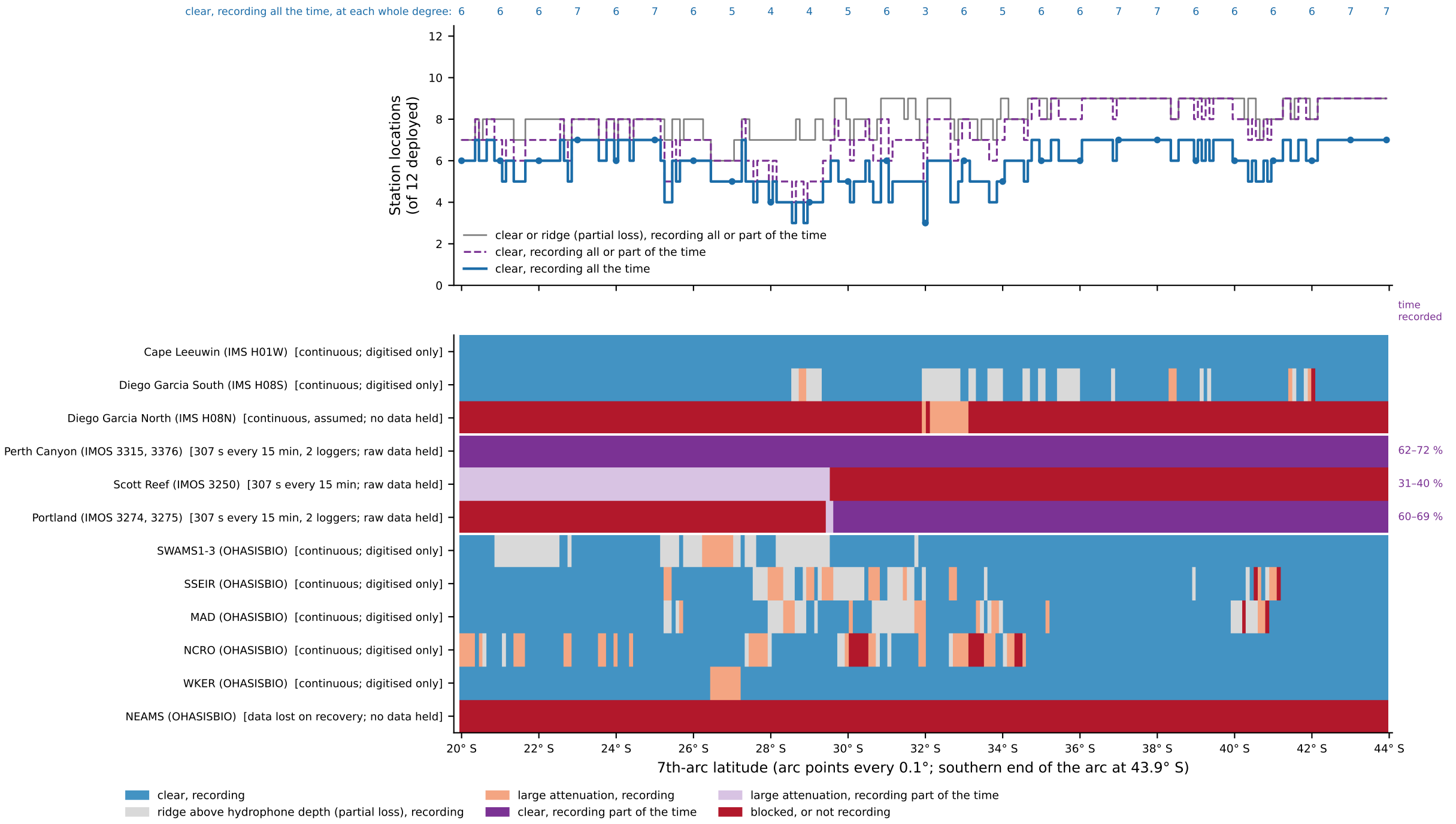
Observation opportunity for a sound from the 7th arc on 8 March 2014: line of sight and recording together.
Recording all the time with clear line of sight: 4–7 station locations south of 33° S, 3–7 north of 33° S; one to two more record part of the time



Plain-language note. For each point on the 7th arc we follow the sea floor to each station, and we look at what the station recorded when the sound would arrive (impact from 00:19:00 to 00:52:00 UTC, plus the travel time). Red: the path is blocked, or the station did not record (NEAMS was recording, but its data was lost when it was recovered). Purple: the Australian seabed loggers record 307 s in each 15 min, so they record only part of the time; the number at the right is the part of the time recorded. This shows where a sound can be received; it does not show that the sound was loud enough to detect. That is the next step.

Technical note. Line of sight: hydro_data.sqlite table arc_path (arc_line_of_sight.py; WGS84 geodesics on GEBCO 2026, 15 arc-s). Blocked: land or water shallower than min(receiver depth, 200 m) more than 20 km from the receiver. Large attenuation: >= 10 km shallower than 1,000 m more than 100 km from the receiver. Ridge: >= 10 km shallower than the hydrophone (deep receivers). These thresholds are agent choices, not ruled. Recording: IMOS record windows from table segment (UTC, clock-corrected, event-day and background sets joined; no record missing 00:00-05:00 UTC); OHASISBIO continuous at 240 Hz (Royer 2016), NEAMS lost during recovery; IMS H01W/H08S continuous 250 Hz (8 Mar 2014 data used in Kadri 2024, CMST 2014); H08N continuous assumed (no published use of its 8 Mar 2014 data found). A location takes its most favourable instrument that is recording. Travel time = geodesic distance / c, c = 1.482 ± 0.006 km/s (registry constant c_sofar); 1-sigma timing error 4-16 s. Not included: HA04 Crozet (installed Dec 2016), IMOS deployments with no 8 Mar 2014 files, island seismometers. Geometric screen, not transmission loss; noise (for example the seismic survey shots Royer reports) is not included. Reception window per arc point: impact 00:19:00-00:52:00 UTC + d/c; 'time recorded' = share of that window covered by the union of the location's unblocked loggers.